

**Universal Station^X
Multiple Schematics
User's Guide**

**Release 300
ML09-300**

Universal Station^x

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Multiple Schematics
User's Guide**

**Release 300
ML09-300
2/99**

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About This Publication

Purpose

This document provides information and procedures on how to install, configure, and use the Multiple Schematics application for a Universal Station^x (U^xS).

Intended audience

Engineers and other technical personnel who will install and configure the Multiple Schematics, and operators and other users of LCN schematics.

Application

This document applies to Multiple Schematics, Release 200, running on TDC 3000^x and requires U^xS related software.

Knowledge required

We presume the following about users of this document:

- People who will install, configure and maintain Multiple Schematics are familiar with:
 - HP-UX and its directory and file structure, UNIX commands, and X-Windows environment.
 - The relationships and interaction between the TDC 3000^x Local Control Network (LCN) and HP-UX in a U^xS.
- People who will use the schematics presented by Multiple Schematics and who will use the standard displays and schematics in the U^xS native window are familiar with:
 - TDC 3000^x process operations functions and procedures.
 - Starting up, general use, and closing of applications in X-Windows.

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Acronyms

CIM	Computer Integrated Manufacturing
IWSM	Industrial Work Space Manager
LAN	Local Area Network
LCN	Local Control Network
OSF	Open Systems Foundation
OSI	Open Systems Interconnection
TDC	Total Distributed Control
TPDG	Turbo Peripheral Display Generator
UCN	Universal Control Network
UNPX	Universal PersonalityX
U ^x S	Universal Station with Extensions
WSI	Workstation Interface

References

Honeywell Documentation

Publication Title	Publication Number	Binder Title	Binder Number
<i>Universal Station^X System Administration Manual</i>	UX11-403	Universal Station ^X	3095
<i>Data Entity Builder Manual</i>	SW11-511	Implementation/Engineering Operations-1	3032-1
<i>Button Configuration Data Entry</i>	SW11-570	Implementation/Engineering Operations-1	3032-1
<i>Button Configuration Form Instruction</i>	SW12-570	Implementation/Engineering Operations-1	3032-1

Hewlett Packard Documentation

Hewlett-Packard CD-ROM Documentation

Hewlett-Packard Documentation at <http://www.docs.hp.com>

1. Introduction

1.1 Multiple Schematics Overview

Product background

The Multiple Schematics application displays TDC 3000^x-based schematics within the X-Windows/HP-UX environment of a Universal Station^x (U^xS). Up to four schematics can be displayed on a local U^xS, or on a remote device (a U^xS, personal computer (PC), or workstation) set up as an X-server and connected to the U^xS through the Plant Information Network (PIN).

Up to four Multiple Schematics windows can be displayed at one time on the local U^xS and on a remote device set up as an X-server.

Features

As illustrated in the following figure, Multiple Schematics allows users to display up to four TDC 3000^x, LCN-based schematics within the X-Windows/HP-UX environment of a U^xS. Other applications can appear in X-Windows on the same screen, at the same time. Both single-screen and dual-screen U^xS,s are supported.

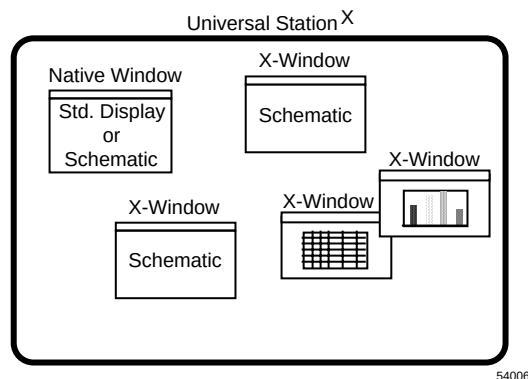


Figure 1-1 U^xS Multiple Schematics Display

Multiple Schematics capability does not sacrifice security or robustness. User-friendly error messages are provided in response to inappropriate user actions and system-related errors.

TDC 3000^x displays

Standard Honeywell displays, such as the System Menu, System Status, Group display, and Detail display are not schematics, and therefore cannot be called up by Multiple Schematics.

Schematic functionality

Multiple Schematics provides functional and visual equivalence for users of TDC 3000^x schematics, including

- keyboard usage,
- custom actors,
- TDC equipment list,
- collectors,
- phantoms,
- overlays,
- trends, and
- cross screen functions.

Performance is similar to that of schematics invoked on a Universal Station (US). The architecture supports porting of Honeywell TDC 3000^x schematic functionality extensions (custom backplane modules) to the Multiple Schematics environment.

Configurable items

Multiple Schematics allows you to configure

- security,
- schematic size,
- location,
- scaling,
- display update rates, and
- LCN resource impact.

Definition of a remote

A remote is defined as an X-server with its own unique internet address. In this manual, a device connected directly to the U^xS by anything other than a PIN is not considered to be a “remote,” and any display appearing on such a device is not considered to be a “remote display.”

Exporting schematics

As illustrated in the following figure, TDC 3000^x schematics can be exported to X-Windows-capable personal computers and workstations connected to a U^xS through the Plant Information Network (PIN).

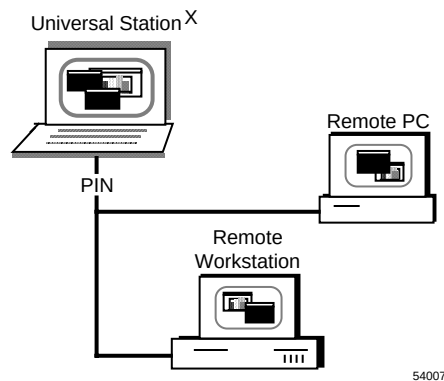


Figure 1-2 Exporting Schematics

Multiple Schematics executes on the X-Window client that exports the schematics to X-Window servers on the remote PCs and workstations.

1.2 Language Support

International language support

Multiple Schematics R200 provides the capability of supporting input text in region-specific local languages. The “internationalize” structure within Multiple Schematics R200 allows regional centers (outside of the U.S.) to implement their native language by developing “localization” modules. Japanese language localization modules have already been developed.

Attention

Each region is responsible for developing its localization modules.

International language support applies to the following items:

- Data point parameters: PTDESC, KEYWORD, EUDESC, STATEn (for HG points), and SO (for PM points).
- Custom Data Segment parameters: String characters, Parameter List identifier, Enumeration identifier, and Enumeration state names.
- NCF definition of the Area Name, and the Area and Console descriptors for USs and U^xSs.
- Run-time picture support for the TEXT, CONDITION, VARIANT, TARGET, and VALUE commands
- Dialog and error messages.

Japanese language users

For Japanese language users, when Multiple Schematics is installed, the following two directories are created:

- /usr/lib/X11/japanese, and
- /usr/lib/X11/japanese/app-defaults.

These directories are created with UNIX permissions of 755. If the directories already exist when Multiple Schematics is installed, the UNIX permissions will be set to 755 by the installation procedure.

1.3 Multiple Schematics Functionality

Multiple Schematics Hosting

As the following illustration shows, the Multiple Schematics application is hosted on the X-side of a U^XS. It exists within the HP-UX and X-Windows environment, getting data and the compiled version of schematics from the US functions in the U^XS and the LCN.

The US's actor interpreter has been ported to the X-side in Multiple Schematics. Thus, the Multiple Schematics application is able to process the action sequence when a target is selected on a schematic.

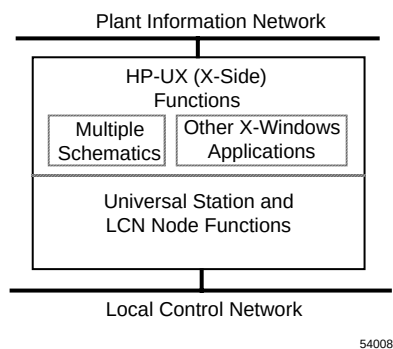


Figure 1-3 Location of Multiple Schematics

Because Multiple Schematics is an X-Windows application, it can export the schematics to other X-Windows-compliant platforms (for example, other workstations, U^XS s, X-terminals, and PCs, hosting an X-server).

Custom backplane modules

The custom backplane modules that you have on your system are still applicable to the current Multiple Schematic release. Backplane modules are loaded using the *update* command. See "Using Update to upgrade other software" in Section 7.8 of the *Universal Station^x System Administration Manual* (UX11-403) for the specific *update* procedure.

Special network connection

In order to use Multiple Schematics locally on a U^xS not connected to a LAN, the U^xS must be connected to a properly terminated Media Access Unit (MAU). The networking software used by Multiple Schematics will not start up correctly unless this connection is made.

Functionality constraints

Here is a summary of the functionality constraints that exist in the initial release of Multiple Schematics, Release 200.

- The capability exists in Release 200 to provide remote stations with both read and write capabilities. Honeywell recommends that you configure them for read only, unless you have specific needs and understand fully the security implications of allowing process parameters to be changed from a remote device.
- The capability also exists in Release 200 to use the Alarm Acknowledge and Alarm Silence Que Keys actors for Operator's keyboard simulation on remote stations. Honeywell recommends that you do not use this functionality unless you have specific needs, and understand fully the security implications of allowing these functions to be executed from a remote device.
- Remote Operator's keyboard functionality will not be supported in Release 200. The absence of such operator keys such as ACK, SCHEM, MAN, SP, and RAMP will affect your ability to control and monitor the process.
- For remotes, Que Key actor support for RAMP keys is not provided.
- RULA capabilities are not supported.

1.4 Technical Assistance

If you need assistance

If you need technical assistance, contact your local Honeywell Service Organization, as explained in the following paragraphs.

International customers

Outside of the United States, contact your local Honeywell Service Organization. If you are not sure of the location or telephone number, call your Honeywell representative for information.

Customers inside the United States

Within the United States, call the Technical Assistance Center (TAC) at the toll free number 1-800-822-7673.

Arizona customers

Within Arizona, the local number for TAC is 602-313-5558.

Services provided

Calls to TAC are answered by a dispatcher from 7:00 A.M. to 5:00 P.M., Mountain Standard Time (6:00 A.M. to 4:00 P.M. when daylight saving time is in effect). Outside of these hours, emergency calls—those which affect your ability to control or view a process—will be received by an answering service, and returned within one hour. TAC maintains its own TDC 3000^x system, and frequently can duplicate problems on this equipment.

Time saving tip

It is a good idea to make specific notes about the problem before making the call. This will help to reduce delays and expedite answers.

2. Installation and Configuration

2.1 Overview

Introduction

This section describes the Multiple Schematics load media and then explains how to install Multiple Schematics, configure the display and security resources, and install PCF fonts.

Model number

To purchase Multiple Schematics, order model number MP-SWXMS4.

Load media

The load media for Multiple Schematics consists of the following two items, packaged separately:

- the Multiple Schematics software, and
- the Multiple Schematics LCN Interface software.

The following table describes the available load media:

Table 2-1 Software Load Media

Part Number	Software Title	Media
51155119	Multiple Schematics	DAT Tape
51155114	Multiple Schematics Load Modules	Bernoulli Cartridge (20 MB)
51155115	Multiple Schematics Load Modules	Bernoulli Cartridge (35 MB)

Multiple Schematics is loaded on a U^xS and the Multiple Schematics Load Modules are loaded on the History Module.

Additional load media

If Honeywell's records indicate that any of your present TDC 3000^x software is below the minimum revision level required for the Multiple Schematics software being provided, additional load media (such as the U^xS Personality, U^xS Windows, etc.) may also be supplied.

Load modules

The Multiple Schematics Load Modules contain the LCN Interface software, which will provide increased independence from future TDC releases. They are loaded as external load modules on your U^xS.

HM loading procedures

The Command Processor is used to load the Multiple Schematics Load Module software on the History Module (HM).

U^xS loading procedures

There are two U^xS loading procedures, depending on whether or not there is a local tape drive on the U^xS where you want the Multiple Schematics software to reside. These two procedures are:

- Local loading procedure
- Remote loading procedure

Local U^xS loading procedure

You use the local U^xS loading procedure to install Multiple Schematics on a U^xS that has a DAT tape drive. You load the Multiple Schematics software directly on to the HP-UX coprocessor's hard disk on the U^xS using the *swinstall* command.

Remote U^xS loading procedure

You use the remote U^xS loading procedure if the U^xS where you want to install Multiple Schematics does not have a DAT tape drive. First of all, you must set up another device (U^xS) with a DAT tape drive as a software distribution depot, and then load the Multiple Schematics software into a directory on this depot using the *swcopy* command. When these tasks are completed, you load the Multiple Schematics software from the depot to the HP-UX coprocessor's hard disk on the U^xS using the *swinstall* command.

***swcopy* command**

The *swcopy* command copies software from a source into a "depot" on the local host. Software that is copied into a depot cannot be used; it must be installed from that depot to make it usable. If your system is to act as a software "source" by other systems, you must first copy the software from the physical media into a depot. *swcopy* can consolidate many different software products and versions into a depot. Configuration is not done with *swcopy*.

***swinstall* command**

The *swinstall* command installs or updates software to the local host from a software source (tape or from a software distribution depot. When you use *swinstall*, software is installed directly into the default directory (/) or into an alternate directory which you specify. *swinstall* automatically configures your system to run the software when it is installed into the default directory.

***update* command**

The *update* command installs or updates software that is not in an *swinstall* format. This would most likely be software supplied by a third party or software that was intended for use on a pre HP-UX 10.20 system. For example, you would use the *update* command to load Multiple Schematics Backplane Modules software. See "Using Update to upgrade other software" in Section 7.8 of the *Universal Station^x System Administration Manual* (UX11-403) for the specific *update* procedure.

Files created during installation

When Multiple Schematics is installed, the following file structure will be created:

```
/opt/TDC_Open/MultSchem/bin/MultSchem
```

Where to learn more

For more information, refer to the HP-UX UNIX "man" pages for *swcopy* and *swinstall*.

Procedure summary

The following table lists the major procedures required to install and configure Multiple Schematics (Mult Schem):

Table 2-2 Procedures to Install and Configure Multiple Schematics

No.	Procedure	Pg.
1	Load Mult Schem Load Modules.	25
2	Configure U ^x S for Load Modules.	27
3	Install Mult Schem to a local node using swinstall.	29
4	Remove software from the software distribution depot, if necessary.	33
5	Set up a software distribution depot using swcopy.	34
6	Install Mult Schem on a remote node using swinstall.	36
7	Configure U ^x S for remote simulation.	41
8	Edit the display resources file.	52
9	Enter display resource values using a Shell command.	54
10	Edit the security resources file.	56
11	Install PCF fonts.	58
12	Add Mult Schem to the workspace menu.	59

2.2 Install Multiple Schematics Load Modules

Overview

The Multiple Schematics Load Modules reside on Bernoulli or zip cartridges as the loadable object files MSCHEM.LO and CSCHEM.LO. The two main procedures for installing the Multiple Schematics Load Modules are:

1. Load the Multiple Schematics Load Modules on the History Module
2. Configure the U^XS for the Load Modules.

Attention

The U^XS where you intend to load Multiple Schematics must already exist as a fully functional node on the LCN. When you add a new U^XS, you must reload all US and U^XS nodes from which the new U^XS is to be accessed to make them visible. Reference the *Implementation/Startup & Reconfiguration -1* binder, *Network Data Entry* manual, Procedure section, "Online Network Reconfiguration" subsection for information on how to add a U^XS to your LCN.

Load Mult Schem Load Modules

This is the first of two consecutive installation procedures. The following table lists the steps required to install the Load Modules "MSCHEM" and "CSCHEM" on the History Module. Execute each step in sequence, using the Auxiliary keyboard on the U^XS.

CAUTION

If you cannot successfully complete any step, do not continue. Call the Honeywell Technical Assistance Center. See Section 1.4, "Technical Assistance," in the Introduction section of this manual.

Table 2-3 Load Mult Schem Load Modules

Step	Action
1	Using the Command Processor of the Engineering or Universal Personality, determine if the directory &CUS exists on the History Module. Enter the command LSV NET The volumes on the network are listed.
2	If the directory does not exist, enter the command CD NET>vol>&CUS where vol is an existing volume on the network. The directory &CUS is created on the History Module.
3	Insert the “Multiple Schematics Load Module” Bernoulli or zip cartridge into the cartridge drive. While the cartridge is being loaded and positioned, both the green and yellow indicator lamps flash. When the green light remains on, the cartridge is ready to use. Note: If you cannot insert the cartridge, carefully remove it and reinsert it. If you are still unsuccessful, contact a maintenance technician for help.
4	Copy the MSCHEM and CSCHEM files from the &MS cartridge to the History Module by entering the following commands: CP \$Fn>&CUS>MSCHEM.LO NET>&CUS>MSCHEM.LO CP \$Fn>&CUS>CSCHEM.LO NET>&CUS>CSCHEM.LO where n is the drive number of the removable media.

Configure U^xS for Load Modules

This is the second of two consecutive installation procedures. The following table lists the steps required to configure the U^xS for MSCHEM and CSCHEM by changing the NCF. Execute each step in sequence, using the Auxiliary keyboard on the U^xS.

CAUTION

If you cannot successfully complete any step, do not continue. Call the Honeywell Technical Assistance Center. See Section 1.4, "Technical Assistance," in the Introduction section of this manual.

Table 2-4 Configure the U^xS for Load Modules

Step	Action
1	Select MODIFY VOLUME PATHS.
2	Set the NCF Backup Path to a removable media pathname by entering the command &Fn>&ASY where n is the number of the drive where you inserted the "Multiple Schematics Load Module" Bernoulli.
3	From the Engineering Personality Main Menu, select LCN NODES. The LCN Node Configuration display appears, with a target for each node on the LCN.
4	Verify that the configurator is on-line. Check the top right corner of the display for the ON-LINE indicator.
5	Select the target for the node number of the U ^x S you want to configure for the Load Modules. Note: You may have to press PAGE FORWARD to get the desired U ^x S on the screen. The Universal Station Node display appears.
6	Select the MODIFY NODE target.
7	On page 2, enter the external load module name MSCHEM and the associated personality type of UP.
8	Enter the external load module name CSCHEM and the associated personality type of UP.

Table 2-4 Configure the U^xS for Load Modules

Step	Action
9	On the line USE DEFAULT PERSONALITY TYPE? select NO. (the default personality is Operator)
10	Select [ENTER]. The configurator calculates the total amount of memory required by reading the header of each custom module file and then adding them up. The total is displayed as TOTAL (MODULES PLUS ADDITIONAL MEMORY) for the UP.
11	Enter the actual amount of memory you want to allocate to the UP in the line MAXIMUM EXTERNAL MODULE MEMORY. To determine this value, use the TOTAL size displayed in the previous step plus a small amount of memory (approximately 2000 is recommended) for an anticipated increase in the size of the custom module. Note: You must enter a value in this field in order to have the software load correctly.
12	Unless the documentation on another of your custom modules states otherwise, FURTHER EXTERNAL DIRECTIVES? should be set to NO for the UP.
13	Select [ENTER].
14	Select [CTL-F1] to check the new NCF, and then press [CTL-F2] to install the new NCF.
15	Select [ENTER] to confirm the installation.
16	Return to the Engineering Personality Main Menu. A message will appear at the bottom of the display indicating that the NCF was successfully installed.
17	Reload the U ^x S. MSCHEM and CSCHEM will be included in the load.

2.3 Installing Multiple Schematics Locally

Required access level

The procedures required to install Multiple schematics on the local U^xS must be performed while logged in as a user in the engineer group. The coprocessor password will be required.

Required hardware

Local installation procedures require a DAT tape unit connected to the U^xS where you want to install Multiple Schematics. The station must be operational with full user login capabilities.

Multiple Schematics DAT tapes

There are two versions of Multiple Schematics for Release 300. Use the tape that is appropriate for your system's LCN release.

- Multiple Schematics 120 on HP-UX 10.20 (R4xx)
- Multiple Schematics 300 on HP-UX 10.20 (R5xx, R6xx)

Install Multiple Schematics locally using *swinstall*

The following table lists the steps required to install Multiple Schematics from a local DAT tape drive. Execute each step in sequence, using the mouse and the Auxiliary keyboard on the local U^xS where you want to install Multiple Schematics.

CAUTION

If you cannot successfully complete any step, do not continue. Call the Honeywell Technical Assistance Center. See Section 1.4, "Technical Assistance," in the Introduction section of this manual.

Table 2-5 Install Multiple Schematics to a Local Node

Step	Action
1	Login as engineer on the U ^X S node.
2	Insert the appropriate Multiple Schematics on HP-UX tape into the local DAT tape drive.
3	On the U ^X S display, select System Menu > Update Software > SWInstall
4	In the Local Software Install window, type the password for root and click OK.
5	A Note (hostname) window may appear indicating that the default source is not valid because you have not yet specified the source. Click OK to continue.
6	A Specify Source (hostname) window appears. Enter the information for the following options: • Source Depot Type . . . – Click to display additional options – Choose Local Tape • Source Host Name . . . <u>hostname of local node</u> • Source Depot Path . . . • Software Filter: <i>none</i> • Click OK.

Table 2-5 Install Multiple Schematics to a Local Node

Step	Action
7	An SD Install - Software Selection (<i>hostname</i>) window appears with a list of products/bundles available on the software distribution tape. • If you wish to install the complete product/ bundle perform the following: – Highlight the software product/bundle. – Select Actions > Install (analysis). • If you wish to view the individual filesets in the software product/bundle, perform the following: – Highlight the software product/bundle and select Actions > Open Item. Note: You may need to perform this step more than one time to display the filesets. – Highlight each individual fileset to be installed. – Use Ctrl + Click to select more than one fileset. – Select Actions > Mark for Install. – Select Actions > Install (analysis).
8	An Error (<i>hostname</i>) window may appear if the selected software has any dependencies. This does not mean, however, that the target location will be unable to satisfy the dependencies - it is just a warning that dependencies do exist. Click OK to continue
9	An Install Analysis (<i>hostname</i>) window appears with the following information displayed: – Target: <i>source host name</i> – Status: <i>Analyzing or Ready</i> – Products Scheduled: <i>software products marked for install</i> Click OK when the Status indicates Ready or Ready with Errors.
10	A Confirmation (<i>hostname</i>) window appears indicating that installation will now begin and that only those products that passed analysis will be installed. Click Yes to continue.

Table 2-5 Install Multiple Schematics to a Local Node

Step	Action
11	An Install (<i>hostname</i>) window appears with the following information displayed: – Target: – Status: – Percent Complete: – Kbytes Installed: – Time Left (minutes): – Loading Software: You may choose to look at the Logfile when the status indicates Completed to determine if there were any errors during installation. To do so, perform the following: – Click on the Logfile button. A Logfile (<i>hostname</i>) window appears. – Scroll through the log to determine if there were any errors during the installation. – When finished looking at the logfile, click OK. – Click Done from the Install Window.
12	Select File > Exit to exit SD Install .
13	If you want to use the Operator's keyboard with Multiple Schematics, check the settings in the following file and make any necessary changes: <code>/usr/lib/X11/usxserverrc</code> For example: Change the following setting: From: -AllowOpKeyboard To: +AllowOpKeyboard
14	• Check the file <code>/etc/services</code> to verify that the port #, XXXXX, in <code>okbasdaemon XXXXX/tcp</code> is unique (typically the next # after <code>lcnd</code>). • If it is not, change the # to a unique value.
15	If your system requires a reboot, perform the following: • Select System Menu > Configuration > System Terminal. • Enter the password for root in the System Terminal Window and click OK. • Type the following at the prompt: <code>/etc/reboot</code>
16	Remove the software distribution tape from the local tape drive.

2.4 Installing Multiple Schematics Remotely

Overview

There are three main procedures for installing Multiple Schematics remotely on a U^xS *without* a DAT tape drive:

1. Remove unneeded software from the software distribution depot, if necessary.
2. Set up a U^xS, one with a DAT tape drive, to be a software distribution depot using *swcopy*.
3. Install Multiple Schematics on the U^xS without a DAT tape drive using *swinstall*.

Required access level

The remote installation procedures must be performed while logged in as “engineer.” The coprocessor password is required.

Required hardware

The loading procedure requires a DAT tape unit connected to the U^xS where you are going to set up the server. The station must be operational with full user login capabilities. This U^xS and the U^xS where you want to install Multiple Schematics must be on the same LAN.

Removing Software from a Software Distribution Depot

During a remote installation, the directory `/var/spool/sw/software` (where *software* is the name of a software subdirectory) is used as an intermediate storage location for software. If this directory already exists on the U^xS where you are going to set up the software distribution depot, it is a good idea to remove the Multiple Schematics software that is no longer required before using the *swcopy* command to load the new software into this depot directory. You will use HP-UX's *swremove* command to accomplish this task. See *Removing Software* in the "Administering Software" section of *Managing Software with SD-UX for HP-UX 10.20 (HP-UX 10.* Operating System)* manual.

Set up a Software Distribution Depot

This is the first of two consecutive remote installation procedures you need to perform. The following table lists the steps required to set up a U^xS with a DAT tape drive to be a software distribution depot, and then load the Multiple Schematics software into a directory on that server depot using the *swcopy* command. Using the mouse and the Auxiliary keyboard on the U^xS with the DAT tape drive, execute each step in sequence unless directed otherwise.

⇒ NOTE: HP Series 700 and Series 800 software cannot coexist in the same depot. If you administer software for both of these platforms, you should create separate depots for each.

CAUTION

If you cannot successfully complete any step, do not continue. Call the Honeywell Technical Assistance Center. See Section 1.4, "Technical Assistance," in the Introduction section of this manual.

Table 2-6 Configure a Software Distribution Depot

Step	Action
1	Log in as engineer to the U ^x S node that will act as the Software Distribution Depot.
2	Insert the Multiple Schematics tape into the local DAT tape drive.
3	On the UXS display, select System Menu > Update Software > SWCopy.
4	In the Create Software Depot window, enter the password for root and click OK.
5	A Select Target Depot Path (<i>hostname</i>) window appears. Enter the following information: • Target Depot Path: <i>/var/spool/sw/software</i> (where <i>software</i> is the subdirectory name) • Click OK.
6	Click OK if a note window appears with the following message: "The default source cannot be used as a source because it is already marked as a target."

Table 2-6 Configure a Software Distribution Depot

Step	Action
7	• A Specify Source (<i>hostname</i>) window appears. Enter the following information: – Source Depot Type: Local Tape – Source Host Name . . . <i>hostname</i> – Source Depot Path: /dev/rmt/0m – Software filter: none • Click OK
8	In the SD Copy - Software Selection (<i>hostname</i>) window, perform the following: • Select Options > Change Options. • A Note (<i>hostname</i>) window appears that recommends using caution when changing options. Click OK to continue. • Uncheck the Enforce dependency analysis errors in agent. • Click OK.
9	In the SD Copy - Software Selection (<i>hostname</i>) window perform the following: • Highlight the appropriate software selection line. • Select Actions > Copy (Analysis).
10	An Error (<i>hostname</i>) window may appear indicating an error because pre-requisite filesets are required. These filesets are already installed on the target U^xS nodes. Press ENTER to continue.
11	Click OK when the Copy Analysis (<i>hostname</i>) window shows a status of Ready with Warnings
12	In the Confirmation (<i>hostname</i>) window, click Yes.
13	A Copy Window (<i>hostname</i>) window appears indicating the following information displayed: – Target: – Status: – Percent Complete: – Kbytes copied: – Time left (minutes): – Copying software: Click Done when the Status indicates completed.

Table 2-6 Configure a Software Distribution Depot

Step	Action
14	Exit SWCopy by selecting File > Exit .
15	Remove the Multiple Schematics tape from the local tape drive.
16	The software is now loaded on the Software Distribution Depot. When you are ready to load the software onto a remote node, go to the next subsection, "Loading the U ^x S Personality from a Software Distribution Depot to a Remote Node."

Install Multiple Schematics to a Remote Node Using *swinstall*

This is the second of two consecutive remote installation procedures you need to perform. The following table lists the steps required to install Multiple Schematics on a U^xS without a DAT tape drive using the *swinstall* command. The installation is performed from the node containing the Software Distribution Depot, which was set up in the previous procedure.

CAUTION

If you cannot successfully complete any step, do not continue. Call the Honeywell Technical Assistance Center. See Section 1.4, "Technical Assistance," in the Introduction section of this manual.

Table 2-7 Install Multiple Schematics on a Remote Node

Step	Action
1	Log in as engineer to the U ^x S node that contains the Multiple Schematic software already loaded on the Software Distribution Depot.
2	On the U ^x S display, select System Menu > Configuration > System Terminal .
3	In the System Terminal window, enter the password for root and click OK if you are not already logged in.
4	Type the following at the prompt: <code>rlogin client_node_name</code> Where <i>client_node_name</i> is the actual name of the node that will have the software installed.
5	Enter the password of the client node and press ENTER.

Table 2-7 Install Multiple Schematics on a Remote Node

Step	Action
6	In the remote System Terminal window, enter the following command: /usr/sbin/swinstall/
7	A screen of instructions appears explaining how to use swinstall through the terminal. Press ENTER to continue.
8	The SD Install - Software Selection (hostname) window appears within the System Terminal Window with a note stating that the software source does not exist. You will define the software source in the next step. Press ENTER to continue.
9	A Specify Source (hostname) window appears with the entry in the Source Depot Type highlighted. Select the appropriate software source and path by performing the following: • Press Enter to display additional options, if necessary. • Use the Up/Down Arrow keys to highlight the software source of Network Directory and press ENTER. • Use the TAB key to go to Source Depot Path. Enter the Target Depot Path defined in the previous procedure. • Press the TAB key to go to OK and then press ENTER.
10	You will be returned to the SD Install - Software Selection (hostname) window. The UXS- PERSONALITY software should be listed in the window. Mark the software for installation by performing the following: • Highlight the software product by pressing the space bar. • Press the TAB key to go to the menu bar. • Use the right arrow key to move to the Actions drop down menu and then press ENTER. • Use the down arrow key to select Install (Analysis) from the Actions menu and then press ENTER.
11	An Error (hostname) window may appear if the selected software has any dependencies. This does not mean, however, that the target location will be unable to satisfy the dependencies - it is just a warning that dependencies do exist. Press ENTER to continue

Table 2-7 Install Multiple Schematics on a Remote Node

Step	Action
12	An Install Analysis (<i>hostname</i>) window appears with the following information displayed: – Target – Status – Products Scheduled The install program is checking for adequate disk space and software dependencies. If all dependencies have been met and there is adequate disk space, the Status will indicate Ready. • If the Status does not indicate Ready, you will need to correct the problem before continuing. • Press the TAB key to go to OK and then press ENTER.
13	A Confirmation (<i>hostname</i>) window appears with the installation information displayed. Press ENTER to continue.
14	An Install Window (<i>hostname</i>) window appears during the installation process with the following information displayed: – Target: <i>hostname</i> – Status: <i>series of status messages</i> – Percent Complete: <i>advances to 100%</i> – Kbytes Installed: – Time Left (minutes): – Loading Software: <i>software product name</i> The Percent Complete will advance to 100%. Do not, however, begin the next step until the Status indicates Completed or Ready.
15	• At this time you can choose to look at the logfile to determine if there were any errors during installation. To do so, perform the following: – Press the TAB key to go to Logfile. – Press ENTER to display all the loaded modules. – A Logfile (<i>hostname</i>) window appears. – Scroll through the log to determine if there were any errors during the installation. • When finished looking at the logfile, press the TAB key to go to OK and then press ENTER.
16	You will be returned to the Install Window (<i>hostname</i>) window. Press the TAB key to go to DONE and then press ENTER.
17	The SD Install - Software Selection (<i>hostname</i>) window appears. • Use the TAB key to go to the File Menu and press ENTER. • Use the down arrow key to go to Exit and press ENTER.

Table 2-7 Install Multiple Schematics on a Remote Node

Step	Action
18	If you want to use the Operator's keyboard with Multiple Schematics, check the settings in the following file and make any necessary changes: <code>/usr/lib/X11/usxserverrc</code> For example: Change the following setting: From: <code>-AllowOpKeyboard</code> To: <code>+AllowOpKeyboard</code>
19	• Check the file <code>/etc/services</code> to verify that the port #, XXXXX, in <code>okbasdaemon XXXXX/tcp</code> is unique (typically the next # after <code>lcmd</code>). • If it is not, change the # to a unique value.
20	If your system requires a reboot, perform the following: • Select System Menu > Configuration > System Terminal. • Enter the password for root in the System Terminal Window and click OK. • Type the following at the prompt: <code>/etc/reboot</code>
21	Remove the software distribution tape from the local tape drive.

2.5 Configure U^xS for Remote Simulation

Overview

Using the Button Configurator, you can configure the programmable function (PF) keys on the U^xS Auxiliary keyboard so that you will be able to simulate an Operator's keyboard from a remote station set up as an X-server on the same LAN. The PF keys (1..12) on the U^xS keyboard map to the F keys (1..12) on an AT style keyboard. For example, when <F1> is pressed on the remote, the action configured for <PF1> on the U^xS will be executed.

There are two ways you can configure a PF key:

1. Use the Que Key actor to emulate almost any key on a U^xS keyboard
2. Specify one or more actors to perform functions such as calling up a display, calling up a schematic, etc.

Configure U^xS for remote simulation

The following table lists the steps required configure a U^xS so you can simulate an Operator's keyboard from a remote station set up as an X-server on the same LAN. Execute each step in sequence unless directed otherwise.

For additional details on this procedure, see the Button Configuration Form Instruction and Button Configuration Data Entry manuals.

CAUTION

If you cannot successfully complete any step, do not continue. Call the Honeywell Technical Assistance Center. See Section 1.4, "Technical Assistance," in the Introduction section of this manual.

Table 2-8 Configure U^xS for Remote Simulation

Step	Action
1	Fill out the Button Configuration forms to document the buttons to be configured and the actions designated for each: For example: To call up a schematic, there are two actors you can use: • SCHEM (R_STR (8, "Enter Schematic Name")) • QUE_KEY (SCHEM)
2	Use the Button Configurator to enter, save, and compile the button configuration data.
3	Label the Operator's keyboard insert to describe the functions set up for each button.
4	Install the new area database.

2.6 Application Resources

Overview

The Multiple Schematics software contains resource files that you can edit to alter the application's behavior to suit your needs. These files reside in the directory `/usr/lib/X11/app-defaults`.

The default resource values are defined later in this section, as part of the description of each resource.

Information on standard X-Window resources and standard HP-UX configuration procedures is provided in “Application Resources” in the HP manual *Using the X-Window System*.

Resource files

There are two application resource files:

- MultSchem
- MultSchem.resources

MultSchem contains the display resources and MultSchem.resources contains the security resources.

Examples of both of these Multiple Schematics resource files follow. Each resource line starts with an “*” followed by the application resource name (matching upper and lower case letters), a colon, a space, and then the value. Comment lines begin the line with a “!”.

MultSchem file

An example of a MultSchem file follows:

```
!
! (this file should exist in the /usr/lib/X11/app-defaults directory)
! (this file provides the configurable display resources and the default access level)
!
!   MultSchem
!
*accessLevel: scViewonly
*focusUpdateRate: 6
*nonfocusUpdateRate: 8
*remoteUpdateRate: 10
*useBackingStore: TRUE
*useScrollBars: FALSE
*abstractName:
*useNativeWindow: TRUE
*nativeFont: -iacd-kanji-medium-r-normal--16-120-100-100-c-160- jisx0208-0
*useDelayActor: TRUE
*allowRemoteAckSil: FALSE
```

MultSchem.resources file

An example of a MultSchem.resources file follows:

```
!
! (this file should exist in the /usr/lib/X11/app-defaults directory)
! (this file contains the client resources that provide security)
!
!   MultSchem.resources
!
*operatorAccess:
*supervisorAccess:
*engineerAccess:
```

Display resources

The Multiple Schematics application understands and utilizes ten application-specific Display resources. They are defined in the MultSchem file. A list of the resources and a definition of each follows:

Table 2-9 Display Resources

Resource Name	Valid Entry
focusUpdateRate	4 (minimum value)
nonfocusUpdateRate	6 (minimum value)
remoteUpdateRate	8 (minimum value)
useBackingStore	TRUE or FALSE
useScrollBars	TRUE or FALSE
abstractName	Abstract Name
useNativeWindow	TRUE or FALSE
nativeFont	Font Name
useNativeFont	TRUE or FALSE
useDelayActor	TRUE or FALSE
allowRemoteAckSil	TRUE or FALSE

focusUpdateRate

This resource specifies the display update rate in seconds for the local display window that *has focus*. (Definition: the window that receives keyboard input is the window that has focus). This resource is only used when the client and the server are the same U^XS. The value is an integer.

Default value: 6

Minimum value: 4

Attention

If a value less than the minimum is specified, the minimum value will be used.

You should consider the trade off between specifying a short update interval, which increases the load on the schematic's LCN "owner" node, and specifying a longer update interval, which may delay the appearance of up-to-the-moment information on the X-side display.

nonfocusUpdateRate

This resource specifies the display update rate in seconds when the display window *does not have focus*. This resource is only used when the client and the server are the same. The value is an integer.

You should consider the trade off between specifying a short update interval, which increases the load on the schematic's LCN "owner" node, and specifying a longer update interval, which may delay the appearance of up-to-the-moment information on the X-side display.

Default value: 8

Minimum value: 6

Attention

If a value less than the minimum is specified, the minimum value will be used.

remoteUpdateRate

This resource specifies the display update rate in seconds for a display on a remote station. This rate is used *whether or not the window has focus*. The value is an integer.

Default value: 10

Minimum value: 8

Attention

If a value less than the minimum is specified, the minimum value will be used.

You should consider the trade off between specifying a short update interval, which increases the load on the schematic's LCN "owner" node, and specifying a longer update interval, which may delay the appearance of up-to-the-moment information on the remote display.

useBackingStore

This resource specifies whether or not a pixmap should be used for drawing the display. The pixmap may impact the update rate, depending on the speed of the X-server, but the screen refresh will be smoother. We suggest that you try using TRUE, and then FALSE. Then choose the value that gives you the best result. The value is Boolean.

Default value: TRUE

useScrollBars

This resource specifies that either (1) a display that exceeds the current window size is clipped to the window so that you use scroll bars to navigate the display, or (2) the display is scaled down to fit the current window. The value is Boolean.

Default value: FALSE (scaled to fit the current window)

Attention

If the Multiple Schematics window is the default size, the scroll bars do not appear until you make the window smaller.

abstractName

This resource specifies the name of the abstract (schematic) to be invoked when a Multiple Schematics (Mult Schem) window is opened for the first time. The value is a character string: the name of any existing schematic that can be found through the Pathname Catalog in the Area Database in use on the Universal Station^x client that is running Multiple Schematics.

Default value: NULL (invoke only the Mult Schem window—no schematic initially appears)

useNativeWindow

This resource specifies that requests for actions that cannot be handled by schematics in X-Windows are either (1) sent to the native window, or (2) return an error message. This resource is only used when the client and server are the same. For remote displays, the error message is always returned. The value is Boolean.

Actions that cannot be handled by schematics in X-Windows are:

- Honeywell standard displays, such as System Summary, System Menu, Alarm Summary, Console Status and Assignment, Group, and Detail displays.
- Cross Screen and From Screen actions
- Actions requested by user-configured buttons
- REMOTE actor
- RULA (Remote User LCN Access) functions

Default value: TRUE (send to native window)

nativeFont

This resource specifies the font to be used when the native language is not English. This font should be the X-Windows equivalent of the LCN font for that language.

Do not change the value of this resource unless you are directed to do so by Honeywell.

Default value: -iacd-kanji-medium-r-normal--16-120- 100-100-c-160-jisx0208-0

useNativeFont

This resource is used in conjunction with the nativeFont resource. This resource should only be set to TRUE if the user requires a non-english font. Since the fonts are scalable, setting this resource to TRUE can greatly increase the time it takes to resize a window if the nativeFont has a large character set. The value is Boolean.

Default value: FALSE

useDelayActor

This resource specifies whether or not to utilize the DELAY actor when executing a target. When the resource is set to FALSE, any actor sequence within a target will execute until encountering the DELAY actor. The DELAY actor will then output an error message, and the remaining actors in the target will not be executed. When the resource is set to TRUE, updating of the schematic—including such objects as values, conditionals, subpictures, and variants—in the Multiple Schematics window is deferred until completion of the delay period. Updating of the schematic in the native window continues during the delay period. The value is Boolean.

Attention

The behavior of a target containing the DELAY actor may seem noticeably different on a Mult Schem window, especially if the delay is for a relatively long period, such as several seconds or more.

Default value: TRUE

allowRemoteAckSil

This resource specifies whether or not the user can acknowledge alarms and silence alarm horns when a Mult Schem window is being displayed on a remote station. When the resource is set to TRUE, pressing the <ACK> target will only acknowledge the alarms for the window that has focus, and only for the points that have ACKSTAT collectors in that window, and pressing the <SIL> target will have the same effect as pressing <SIL> at the station from which the delay is exported.

This resource is used *only* when the display is exported with control, that is, when the accessLevel is *not* scViewonly. The value is Boolean.

Default value: FALSE

Security resources

The Multiple Schematics application can utilize four application-specific security resources. They are defined in the MultSchem and MultSchem.resources files. A table listing these resources, then a definition of each follows:

Table 2-10 Security Resources

Resource Name	Valid Entry
accessLevel	scViewonly, or scOperator, or scSupervisor, or scEngineer
operatorAccess	Server ID
supervisorAccess	Server ID
engineerAccess	Server ID

accessLevel

This resource specifies the maximum level for which the user has authorization. This resource only applies when the display is on a remote station. The possible values are view-only, operator, supervisor, and engineer.

Default value: scViewonly

Valid entries: scViewonly
scOperator
scSupervisor
scEngineer

This resource resides on remote servers.

See *Remote server security mechanism* in Section 3.5, "Security."

operatorAccess

This resource specifies the list of remote X-servers that are allowed to display Multiple Schematics when the keylevel on the client U^xS is operator or view-only.

The intention is that for a subsequent release of Multiple Schematics, this resource will imply that this server will execute the Multiple Schematics application at this security level, regardless of the keylevel of the local U^xS.

Default value: " "

This resource resides on the client, not on any server.

supervisorAccess

This resource specifies the list of remote X-servers that are allowed to display Multiple Schematics when the keylevel on the client U^xS is supervisor, operator or view-only.

The intention is that for a subsequent release of Multiple Schematics, this resource will imply that this server will execute the Multiple Schematics application at this security level, regardless of the keylevel of the local U^xS.

Default value: " "

This resource resides on the client, not on any server.

engineerAccess

This resource specifies the list of remote X-servers that are allowed to display Multiple Schematics when the keylevel on the client U^xS is in any position.

The intention is that for a subsequent release of Multiple Schematics, this resource will imply that this server will execute the Multiple Schematics application at this security level, regardless of the keylevel of the local U^xS.

Default value: " "

This resource resides on the client, not on any server.

Xt resources

The Multiple Schematics application also understands and utilizes three Xt resources that can appear on the command line. They are described in the following table:

Table 2-11 Xt Resources

Resource Name	Valid Entry
display	Server to use
geometry	Size and Location
iconic	No value required

display

This is a standard command line option that specifies where the X-Window is to be displayed.

geometry

This is a standard command line option that specifies the X-Window's screen size and location.

iconic

This is a standard command line option used to start an X-Window as an icon.

2.7 Configure Display Resources

Overview

There are two ways to configure display resources:

1. Edit the MultSchem configuration file.
2. Enter resource values from a Shell command line.

Edit the display resources file

The MultSchem configuration file for display resources resides in the /usr/lib/X11/app-defaults directory. The display resource values in this file apply to all schematics that are displayed. You can, however, override display resource values for a specific schematic by executing a Shell command.

The following table lists the steps required to enter new display resource values into the MultSchem configuration file. Use the mouse and the Auxiliary keyboard on the U^xS where Multiple Schematics is installed. Execute each step in sequence unless directed otherwise.

CAUTION

If you cannot successfully complete any step, do not continue. Call the Honeywell Technical Assistance Center. See Section 1.4, "Technical Assistance," in the Introduction section of this manual.

Table 2-12 Edit Display Resource Files

Step	Action
1	Log in as “engineer” from the Login banner at the bottom of the screen by executing the following steps: • Position the cursor in the Login: input port, type in your user name, and then press the <RETURN> key on the keyboard. The Password: input port will automatically be selected. • Type in your password and press <RETURN>. The native LCN window is reduced to 1/4 size and appears in the top left corner of the screen, and the bpicker window, indicating which mouse button is active, appears in the bottom right corner.
2	Position the cursor anywhere in the workspace and press the mouse button. The Engineering Menu appears. Without releasing the button, make the following three selections: • Select System Menu. The System Commands menu appears. • Select Configuration. The Configuration menu appears. • Select System Terminal.... The System Terminal window appears.
3	Position the cursor in the Password: input port, type in the required password, and then press <RETURN>. The System_Terminal window appears, with the text input port at the end of the last line.
4	To enter new display resource values in the MultSchem file using the vi editor, proceed as follows: • Enter the command: cd /usr/lib/X11/app-defaults • Enter the command: vi MultSchem • Position the cursor on the first character of the resource value you want to change using the arrow keys on the keyboard. • Enter the command: cw • Enter the new value. • Press <Esc>. • Repeat the last four actions starting with the cursor positioning for each additional resource value you want to change. • Enter the command :wq! The changes are saved and you exit the vi editor.
5	Position the cursor in the target in the top left corner of the window and double-click the mouse. The System_Terminal window disappears.

Enter display resource values using a Shell command

The following table lists the steps required to enter new display resource values for a specific display from a Shell command line. Resources are made available to the Multiple Schematics application as Shell command line parameter options by preceding the resource with a dash (-). Resource values specified from a Shell command line override the configuration file values. Use the mouse and the Auxiliary keyboard on the U^xS where Multiple Schematics is installed. Execute each step in sequence unless directed otherwise.

CAUTION

If you cannot successfully complete any step, do not continue. Call the Honeywell Technical Assistance Center. See Section 1.4, "Technical Assistance," in the Introduction section of this manual.

Table 2-13 Enter Display Resource Values from a Shell Command

Step	Action
1	Log in as "engineer" from the Login banner at the bottom of the screen by executing the following steps: - Position the cursor in the Login: input port, type in your user name, and then press the <RETURN> key on the keyboard. The Password: input port will automatically be selected. - Type in your password and press <RETURN>. The native LCN window is reduced to 1/4 size and appears in the top left corner of the screen, and the bpicker window, indicating which mouse button is active, appears in the bottom right corner.
2	Position the cursor anywhere in the workspace and press the mouse button. The Engineering Menu appears. Without releasing the button, make the following three selections: - Select System Menu. The System Commands menu appears. - Select Configuration. The Configuration menu appears. - Select System Terminal.... The System Terminal window appears.
3	Position the cursor in the Password: text input port, type in the required password, and press the <RETURN> key. The System_Terminal window appears, with the text input port at the end of the last line.
4	Type in the desired Shell command that includes one or more parameter options separated by a single space, then press <RETURN>. The new display resource value is entered. Note: If the abstractName parameter option is included, a Mult Schem window containing the specified schematic will appear.

Shell command line examples

Following is a list of Shell command line examples for the display resources in the MultSchem configuration file. The Honeywell-provided display PERFMENU is used in the examples. PERFMENU was developed and compiled with the Picture Editor, and is, therefore, a schematic display. Additional Honeywell schematic displays provided with this package include all of the displays called from PERFMENU, \$LNMENU, and all of the displays called from \$LNMENU.

Table 2-14 Shell Command Line Examples

Display Resource	Shell Command Line
focusUpdateRate	MultSchem -focusUpdateRate 5 -abstractName PERFMENU
nonfocusUpdateRate	MultSchem -nonfocusUpdateRate 7 -abstractName PERFMENU
remoteUpdateRate	MultSchem -remoteUpdateRate 9 -abstractName PERFMENU
useBackingStore	MultSchem -useBackingStore FALSE -abstractName PERFMENU
useScrollBars	MultSchem -useScrollBars TRUE -abstractName PERFMENU
abstractName	MultSchem -abstractName PERFMENU
useNativeWindow	MultSchem -useNativeWindow FALSE -abstractName PERFMENU
nativeFont	MultSchem -abstractName PERFMENU -nativeFont -iacd-kanji- medium-r-normal--16-120-100-100-c-160-jisx0208-0
useNativeFont	MultSchem -useNativeFont FALSE -abstractName PERFMENU
useDelayActor	MultSchem -useDelayActor FALSE -abstractName PERFMENU
allowRemoteAckSil	MultSchem -allowRemoteAckSil TRUE -abstractName PERFMENU

2.8 Configure Security Resources

Edit the security resources file

This procedure is optional. Security resources are configured by editing MultSchem.resources, the security resource configuration file, which resides in the /usr/lib/X11/app-defaults directory. The security resource values in this file apply to all schematics that are displayed.

CAUTION

For the initial release of Multiple Schematics we strongly recommend that you **do not** configure remote stations for operatorAccess, supervisorAccess, or engineerAccess.

In a future release, Honeywell intends to develop security mechanisms that will make the input of data from remote servers, to be passed on to LCN nodes, safe. Until these security mechanisms are in place, the consequences of accepting data from remote servers are the users' sole responsibility.

The following table lists the steps required to enter new security resource values from the Shell command line. Use the mouse and the Auxiliary keyboard on the U^xS where Multiple Schematics is installed. Execute each step in sequence unless directed otherwise.

CAUTION

If you cannot successfully complete any step, do not continue. Call the Honeywell Technical Assistance Center. See Section 1.4, "Technical Assistance," in the Introduction section of this manual.

Table 2-15 Edit Security Resource Files

Step	Action
1	Log in as “engineer” from the Login banner at the bottom of the screen by executing the following steps: • Position the cursor in the Login: input port, type in your user name, and then press the <RETURN> key on the keyboard. The Password: input port will automatically be selected. • Type in your password and press <RETURN>. The native LCN window is reduced to 1/4 size and appears in the top left corner of the screen, and the bpicker window, indicating which mouse button is active, appears in the bottom right corner.
2	Position the cursor anywhere in the workspace and press the mouse button. The Engineering Menu appears. Without releasing the button, make the following three selections: • Select System Menu. The System Commands menu appears. • Select Configuration. The Configuration menu appears. • Select System Terminal.... The System Terminal window appears.
3	Position the cursor in the Password: input port, type in the required password, and press the <RETURN> key. The System_Terminal window appears, with the text input port at the end of the last line.
4	To enter remote nodes in the MultSchem.resources file using the vi editor, proceed as follows: • Enter the command: cd /usr/lib/X11/app-defaults • Enter the command: vi MultSchem.resources • Position the cursor on the access level list where you want to enter a remote node using the arrow keys on the keyboard. • Enter the command: <shift>a • Enter the new value(s), with each value separated by a single space if you are entering more than one value. • Repeat the last three actions starting with the cursor positioning for each additional remote node you want to enter. • Press <Esc>. • Enter the command :wq! The changes are saved and you exit the vi editor.
5	Position the cursor in the target in the top left corner of the window and double-click the mouse. The System_Terminal window disappears.

2.9 Install Fonts

Overview

Multiple Schematics allows you to display TDC 3000^x schematics on remote PCs and workstations that are X-Windows compliant. In order to display the schematics precisely, correct fonts must be installed.

PCF fonts

The following PCF (Portable Compiled Format) fonts were loaded into the /usr/lib/X11/fonts/misc directory when you installed the Multiple Schematics package:

- full.pcf
- half.pcf
- kanji.pcf
- lcn8x16.pcf
- lcn8x8.pcf

Installing these five PCF fonts on the remote PC or workstation where you want the schematics to appear insures that they will be displayed precisely.

Install PCF fonts

This procedure is optional, but recommended. Perform the following steps for each remote PC and workstation where you want schematics to be displayed:

Attention

Refer to the font installation instructions for the X-server on the PC or workstation for details (such as the name of the font directory, specific font installation instructions, etc.) on how to perform the following steps

Table 2-16 Install PCF Fonts

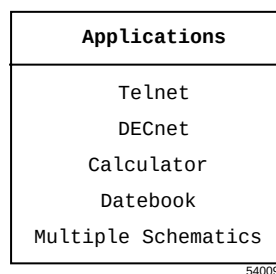
Step	Action
1	Transfer the five PCF font files listed above to the desired X-Windows compliant PC or workstation using an appropriate application, such as FTP, in the ASCII mode.
2	Compile these fonts into the X-server's font database.

2.10 Add Multiple Schematics to the Workspace Menu

Overview

The workspace menu is used to start up a software application simply by selecting a menu target. By editing the appropriate files, you can add an entry to the workspace menu to automatically invoke Multiple Schematics. This is, however, an optional procedure. Reference the *Universal Station^x System Administration Manual*, section on “Customizing the Workspace Menu” for additional information.

If you add Multiple Schematics to the Applications workspace menu, it will appear as follows:



Access groups

You can specify which access groups will be permitted to invoke Multiple Schematics. There are four access groups, which are:

- Operator,
- Supervisor,
- Engineer, and
- ViewOnly.

Users

You can also specify that the Multiple Schematics menu target appears on the workspace menu on a U^xS for

- a current user, by modifying the .mwsrc file for the desired access group in the user's home directory, or for
- all future users, by modifying the .mwsrc file for the desired access group in the .baseuser directory.

Attention

The .mwsrc file must be modified on each U^xS where you want the Multiple Schematics target to appear.

Add Mult Schem to the workspace menu

The following table lists the steps required to add Multiple Schematics to the workspace menu by editing the appropriate files. Execute each step in sequence unless directed otherwise.

Table 2-17 Add Mult Schem to the Workspace Menu

Step	Action
1	Log in as “engineer” from the Login banner at the bottom of the screen by executing the following steps: - Position the cursor in the Login: input port, type in your user name, and then press the <RETURN> key on the keyboard. The Password: input port will automatically be selected. - Type in your password and press <RETURN>. The native LCN window is reduced to 1/4 size and appears in the top left corner of the screen, and the bpicker window, indicating which mouse button is active, appears in the bottom right corner.
2	Position the cursor anywhere in the workspace and press the mouse button. The Engineering Menu appears. Without releasing the button, make the following three selections: - Select System Menu. The System Commands menu appears. - Select Configuration. The Configuration menu appears. - Select System Terminal.... The System Terminal window appears.
3	Position the cursor in the Password: input port, type in the required password, and then press <RETURN>. The System_Terminal window appears, with the text input port at the end of the last line.
4	To provide the Multiple Schematics menu target for a current user, enter one of the following four commands to specify the user’s home directory for the access level you want to assign: <pre>cd /users/XUS/operator/username cd /users/XUS/supervisor/username cd /users/XUS/engineer/username cd /users/XUS/viewOnly/username</pre> where username = user’s login name To provide the Multiple Schematics menu target for all future users, enter one of the following commands to specify the .baseUser directory for the access level you want to assign: <pre>cd /users/XUS/operator/baseUser cd /users/XUS/supervisor/baseUser cd /users/XUS/engineer/baseUser cd /users/XUS/viewOnly/baseUser</pre>

Table 2-17 Add Mult Schem to the Workspace Menu

Step	Action
5	Modify the .mwsrc file using the vi editor by proceeding as follows: - Enter the following command: vi .mwsrc. - Add the lines shown in bold print to the Menu ApplicationMenu section of the .mwsrc file as shown below: <pre>Menu ApplicationMenu { "Applications" f.title "Telnet" f.exec "xterm -title Telnet -ls -geom +291+400 -e telnet &" "DECnet" f.exec "xteemx340 -title DECnet -geom +291+400 &" "Calculator" f.exec "xcalc -geom -160+4" "Datebook" f.exec "datebook -geom +724+4 &" "Multiple Schematics" f.exec "/opt /TDC_Open/MultSchem/bin/MultSchem -display XXXXXXXX:0.0 -abstractName NNNNNNNN &" }</pre> where XXXXXXX is the IP host name (or a valid alias) of the U^xS and NNNNNNNN is the name of the schematic you want to appear. - Press <Esc>. - Enter the command :wq! The changes are saved and you exit the vi editor.
6	If you want to provide the Multiple Schematics menu target for another single user, repeat steps 4 and 5.

3. Operation

3.1 Overview

Background

The Universal Station was developed as a console interface for a single user. Multiple Schematics can, however, be accessed locally, at a U^xS, or remotely, through an X-server. The ability to export schematics to other stations is a very powerful capability which extends the LCN interface to multiple users.

Local operation

Multiple Schematics provides a local user with a great expansion of capability by allowing the user to call up and interact with up to four schematics at the same time (as shown in the following figure). The user has virtually identical capabilities to those possible through schematics displayed on a US or in the native window of the U^xS.

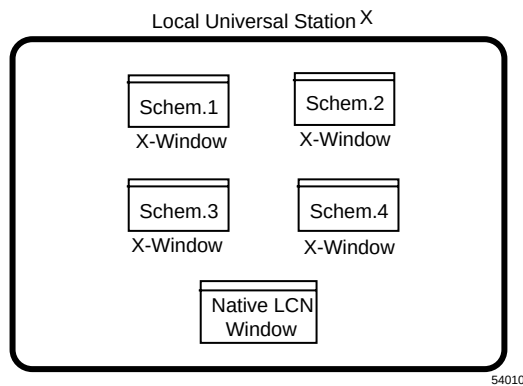


Figure 3-1 Default Configuration for Multiple Schematics

Remote operation

Although an Operator's keyboard is not available when Multiple Schematics is exported to an X-server on a PC or a workstation, Operator's keyboard functionality can be simulated at a remote station. To call up a Mult Schem display at a remote station, you should call for a schematic that contains targets you can use to call up any other schematics you may need, unless you have simulated Operator's keyboard functionality on the remote. See Section 2.5, "Configure U^xS for Remote Simulation."

Interacting with schematics on the X-side

From the local U^xS, your interaction with a schematic is virtually identical to what it would be with a schematic in the native window. You can select targets, enter values, and make normal use of the Operator's keyboard.

From a remote station, you can select targets, but in the initial release of Multiple Schematics, Operator's keyboard inputs other than alphanumerics and <RETURN> are not accepted from remote X-servers.

Interacting with the native LCN window

If you have an Engineer's display such as the Engineering Main Menu or the Command Processor in the native LCN window and you request a function on the Operator's keyboard such as GROUP or DETAIL, the function will not work. If, however, you have an Operator's display such as a schematic in the native LCN window, an Operator's keyboard request will work. For example, pressing the DETAIL key will open up a text input port in the native LCN window asking for the group number.

Standard Honeywell displays

Schematics running on the X-side may contain targets that call for standard Honeywell displays such as the System Status display, a Group display, or a Detail display. These displays however cannot appear in a Mult Schem X-Window.

Procedure summary

The following table lists the procedures an operator can perform.

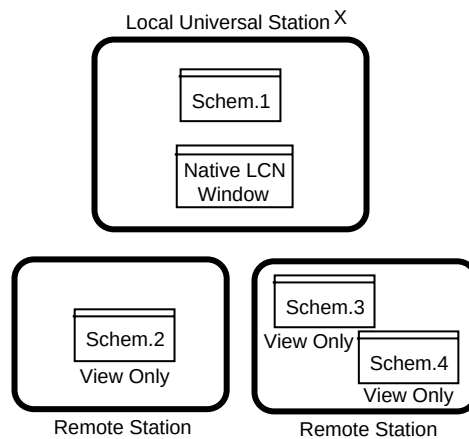
Table 3-1 Operator Procedures

#	Procedure
1	Display a window and a schematic locally by a Shell command.
2	Display a window and a schematic locally from a menu.
3	Display a schematic locally.
4	Display a window and a schematic remotely.
5	Display a schematic remotely.

3.2 Remote Station Functionality

View only at remote stations

In the initial release of Multiple Schematics, the primary focus for remote usage of exported schematics is view only. This is the default configuration for Multiple Schematics and allows users at trusted stations to view LCN schematics, as shown in the following figure.



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Read/Write at remote stations

In the future, this type of usage will be very powerful. It will allow customers to distribute the process user interface to stations throughout a plant and business, allowing the operator, engineer, supervisor, and others to have access to the same information.

CAUTION

Read/Write remote usage is recommended only in certain controlled situations for users who recognize and accept certain near term functionality constraints and risks. Multiple Schematics allows multiple windows and stations to be populated with schematics, but the basic design is for a single user, as it relates to security and operational side-effects.

The person who configures the system must determine which stations and which users on those stations have access to the LCN through this interface. By default, even trusted remote users (users who have HP-UX and network access privileges) are granted view-only privileges.

In later releases of the U^xS and Multiple Schematics, Honeywell intends to change the single user design of the U^xS to include multiple users.

3.3 Operate Multiple Schematics Locally

Overview

There are two ways to display a Multiple Schematics X-Window (Mult Schem window) and a schematic locally on the U^xS where Multiple Schematics is installed:

1. Enter a Shell command.
2. Select a workspace menu target that has been configured to perform those functions. Reference 2.10 Add Multiple Schematics to the Workspace Menu.

Display a window and a schematic locally by a Shell command

The following table lists the steps required to call up a new Mult Schem window and display a new schematic from a Shell command line. Use the mouse and the Auxiliary keyboard on the local U^xS. Execute each step in sequence unless directed otherwise.

CAUTION

If you cannot successfully complete any step, do not continue. Call the Honeywell Technical Assistance Center. See Section 1.4, "Technical Assistance," in the Introduction section of this manual.

Table 3-2 Display a Window and Schematic by a Shell Command

Step	Action
1	Log in as “operator” from the Login banner at the bottom of the screen by executing the following steps: • Position the cursor in the Login: input port, type in your user name, and then press the <RETURN> key on the keyboard. The Password: input port will automatically be selected. • Type in your password and press <RETURN>. The native LCN window is reduced to 1/4 size and appears in the top left corner of the screen, and the bpicker window, indicating which mouse button is active, appears in the bottom right corner.

Table 3-2 Display a Window and Schematic by a Shell Command

Step	Action
2	Position the cursor anywhere in the workspace and press the mouse button. The Operator Menu appears. Without releasing the button, make the following two selections: • Select Application Menu. The Applications menu appears. • Select Telnet. The Telnet window appears.
3	Enter the command open XXXXXXXX (XXXXXXXX = IP host name (or a valid alias) of the local U ^x S). Then press <RETURN>. The Login: input port appears.
4	Type in your user name, and then press <RETURN>. The Password: input port appears.
5	Type in your password, and then press <RETURN>. The text input port for a Shell command line appears.
6	Enter the command: MultSchem -display XXXXXXXX:0.0 -abstractName NNNNNNNN & (XXXXXXXX = IP host name of the local U ^x S) (NNNNNNNN = name of the schematic). Then press <RETURN>. A Mult Schem window containing the requested schematic appears on the local U ^x S.

Display a window and a schematic locally from a menu

The following table lists the steps required to call up a new Mult Schem window and display a new schematic from a workspace menu previously configured. Use the mouse and the Auxiliary keyboard on the local U^xS. Execute each step in sequence unless directed otherwise.

CAUTION

If you cannot successfully complete any step, do not continue. Call the Honeywell Technical Assistance Center. See Section 1.4, "Technical Assistance," in the Introduction section of this manual.

Table 3-3 Display a Window and a Schematic Locally from a Menu

Step	Action
1	Log in as “operator” from the Login banner at the bottom of the screen by executing the following steps: • Position the cursor in the Login: input port, type in your user name, and then press the <RETURN> key on the keyboard. The Password: input port will automatically be selected. • Type in your password and press <RETURN>. The native LCN window is reduced to 1/4 size and appears in the top left corner of the screen, and the bpicker window, indicating which mouse button is active, appears in the bottom right corner.
2	Position the cursor anywhere in the workspace and press the mouse button. The Operator Menu appears. Without releasing the button, make the following two selections: • Select Application Menu. The Applications menu appears. • Select Multiple Schematics. A Mult Schem window appears, displaying the schematic that was set up. Reference: Section 2.10, "Add Multiple Schematics to the Workspace Menu."

Display a schematic locally

The following table lists the steps required to display a new schematic in an *existing* Mult Schem window on the local U^xS. Execute each step in sequence unless directed otherwise, using the mouse and the Auxiliary keyboard on the local U^xS.

CAUTION

If you cannot successfully complete any step, do not continue. Call the Honeywell Technical Assistance Center. See Section 1.4, "Technical Assistance," in the Introduction section of this manual.

Table 3-4 Display a Schematic Locally

Step	Action
1	Select the desired Mult Schem window as the focus window.
2	If the schematic in this Mult Schem window contains a target configured to call up another schematic, you may select it instead of executing the remaining two steps, and the new schematic will replace the one in the window.
3	Press <SCHEM> on the Operator's keyboard. A text input port appears in the Mult Schem window. Note: If the text input port appears in the native window, not in the Mult Schem window, check to see that +AllowOpKeyboard setting appears in the /usr/lib/X11/usxserverrc file. Reference: Install Multiple Schematics locally using <i>swinstall</i>
4	Type in the name of the desired schematic and press <ENTER>. The new schematic is displayed in the Mult Schem window. Note: If there was a schematic originally on display, it will be overlaid.

Standard Honeywell displays

Standard Honeywell displays *cannot* appear in a Mult Schem window. When a local request for a standard display is made through a target in Mult Schem window, either

- The display appears in the *native window*, or
- The requested display does not appear, but an error message does appear in the Mult Schem window,

depending on the value configured for the `useNativeWindow` display resource. See *useNativeWindow* in Section 2.6, "Application Resource."

3.4 Operate Multiple Schematics Remotely

Overview

Initially, there are two main procedures that must be executed to display a schematic on an X-server:

1. Bring up a Multiple Schematics (Mult Schem) X-Window on the server.
2. Display the schematic in the Mult Schem window.

For local operation, when the Mult Schem window is being displayed on the same U^xS where Multiple Schematics is running (the same device is both server and client), you can use the <SCHEM> button on the Operator's keyboard to display a schematic.

For remote operation, if the remote device is a U^xS, the <SCHEM> button cannot be used. If the remote device is not a U^xS, a PC for example, there is no <SCHEM> button. In both of these cases, you need to configure the programmable function (PF) keys on the U^xS to perform the same function as the <SCHEM> button on the Operator's keyboard.

Prerequisite: Configure U^xS for a remote display

Using the second page of the Button Configurator, you can specify what action to take when a programmable function key on a remote X-server is pressed. For example, if PF11 on the U^xS is configured to execute the SCHEM actor, when you press <F11> on the remote you will be able to enter the name of the schematic you want to display on the remote. This button configuration has to be performed before you can display a new schematic on the screen of remote X-server from the remote device. See Section 2.5, "Configure U^xS for Remote Simulation," for additional information. All remote servers assume this button configuration, which is loaded in the area database in the client U^xS.

Display a window and a schematic remotely

The following table lists the steps required to display a Mult Schem window and a schematic on a remote X-server. Execute each step in sequence unless directed otherwise.

CAUTION

If you cannot successfully complete any step, do not continue. Call the Honeywell Technical Assistance Center. See Section 1.4, "Technical Assistance," in the Introduction section of this manual.

Table 3-5 Display a Window and a Schematic Remotely

Step	Action
1	Check the remote device where you want to display the schematic. Get the X-server up and running if it is not currently in that state.
2	Using the mouse and the Auxiliary keyboard on the U ^x S where Multiple Schematics is installed, log in as "operator" by executing the following steps: • Position the cursor in the Login: input port at the bottom of the screen, type in your user name, and then press the <RETURN> key on the keyboard. The Password: input port will automatically be selected. • Type in your password and press <RETURN>. The native LCN window is reduced to 1/4 size and appears in the top left corner of the screen, and the bpicker window, indicating which mouse button is active, appears in the bottom right corner.
3	Position the cursor anywhere in the workspace and press the mouse button. The Operator Menu appears. Without releasing the button, make the following two selections: • Select Application Menu. The Applications menu appears. • Select Telnet. The Telnet window appears.
4	Enter the command: open XXXXXXXX (XXXXXXXX = IP host name of the U ^x S you are using). Then press <RETURN>. The Login: input port appears.
5	Type in your user name, and then press <RETURN>. The Password: input port appears.

Table 3-5 Display a Window and a Schematic Remotely

Step	Action
6	Type in your password, and then press <RETURN>. The text input port for a Shell command line appears.
7	Enter the command: MultSchem -display YYYYYYYYY:0.0 -abstractName NNNNNNNN & (YYYYYYYYY = IP host name of the remote) (NNNNNNNN = name of the schematic). Then press <RETURN>. A Mult Schem window containing the requested schematic appears on the remote.

Display a schematic remotely

The following table lists the steps required to display a new schematic in an *existing* Mult Schem window on a remote X-server from the remote device. Execute each step in sequence unless directed otherwise, using the mouse and the keyboard on the remote device.

CAUTION

If you cannot successfully complete any step, do not continue. Call the Honeywell Technical Assistance Center. See Section 1.4, "Technical Assistance," in the Introduction section of this manual.

Table 3-6 Display a Schematic Remotely

Step	Action
1	Select the desired Mult Schem window as the focus window.
2	Press the F function key previously configured to call up a schematic. The ENTER SCHEMATIC NAME text input port appears. Reference: Configure U ^x S for a remote display.
3	Enter the name of the desired schematic. The new schematic is displayed in the Mult Schem window on the remote. Note: If there was a schematic originally on display, it will be overlaid.

Data inputs

Honeywell does not recommend that remote X-servers be configured to enable data to be sent through to LCN nodes. Data inputs from a remote station should be view-only, by default.

Standard Honeywell displays

Standard Honeywell displays *cannot* appear in a Mult Schem window. When a request for a standard display is made through a target in Mult Schem window on a remote device, an error message appears.

3.5 Security

X-side security policy

The security provided by Multiple Schematics is meant to augment the basic security provided by the HP-UX operating system and X-Windows. HP-UX provides user id and password protection which, if used judiciously, provides a reasonable level of protection. However, if this security is compromised, the additional security provided by Multiple Schematics cannot guarantee protection against unauthorized access to the LCN via Multiple Schematics.

Additionally, users can increase the level of network security by using higher-security network topologies. As an example, a user may segment the network into general purpose backbones with hardware-secured segments. The hardware can physically restrict outside access to those segments. The nodes within those segments are trusted and can have freer, less restrictive interaction.

U^xS security mechanism

The U^xS uses two mechanisms to augment the HP-UX security:

- Network security
- X-Windows

Network security is provided by the U^xS keyswitch. The keyswitch level approach to security is extended to include Multiple Schematics and is shared among **all** of the Mult Schem windows, **local and remote**, and among native window displays appearing on a U^xS.

The second complementary security mechanism is the set of configurable resources located on the U^xS which allows the user at a U^xS to specify the level of access for remote displays. The access level that the remote server is requesting is compared against the defined level of access that the U^xS is allowing that server.

The access levels that can be defined for a remote display are:

- Viewonly (default level), (scViewonly)
- Operator, (scOperator)
- Supervisor, and (scSupervisor)
- Engineer (scEngineer)

These access levels are located in the *MultSchem file* on the U^xS. See *MultiSchem file* in Section 2.6, "Application Resources."

Remote server security mechanism

Remote server security is built around access levels. An access level must be assigned to each Multiple Schematic display request initiated from a remote server.

Access level determination

The assigned access level is determined by checking the following three things:

1. The remote user's input on a command line.
2. The access level for the U^XS configured in the MultSchem file.
3. An optional entry in \$HOME.Xdefaults file that a user can set up when logging in on the remote.

The checking logic gives the command line input the highest priority, the optional \$HOME.Xdefaults file entry the next highest, and the MultSchem file entry the lowest.

In the following table, the shaded block identifies the assigned access level finally given to a remote request. For example, the first line of the table shows that when an access level is specified on the command line input from a remote, it becomes the assigned access level, regardless of any other access levels that may or may not be specified in either the \$HOME.Xdefaults or MultSchem files.

Table 3-7 Assigning an Access Level

Access Level Sources		
Command Line Input	\$HOME.Xdefaults File	MultSchem File
Entry	Entry or No Entry	Entry or No Entry
No Entry	Entry	Entry or No Entry
No Entry	No Entry	Entry

Remote server security logic

Remote server security logic can be divided into two main functions.

1. When the client (U^XS) receives a display request from a remote server, the first thing Multiple Schematics does is check the access level lists located in its MultSchem.resources file to determine where the remote is entered. The requested access level must be the same as the access level list where the remote is entered. If it is, the first “security hurdle” is passed. If it is not, the request is denied, unless the requested access level is scViewonly, in which case the request is permitted.

For example, if remote X-Server “ws01” is requesting Operator access, “ws01” must be entered in the operatorAccess level list on the local U^XS.

“ws01” could, of course, be configured for access levels other than Operator. This would allow different users to use that station and request different levels of access.

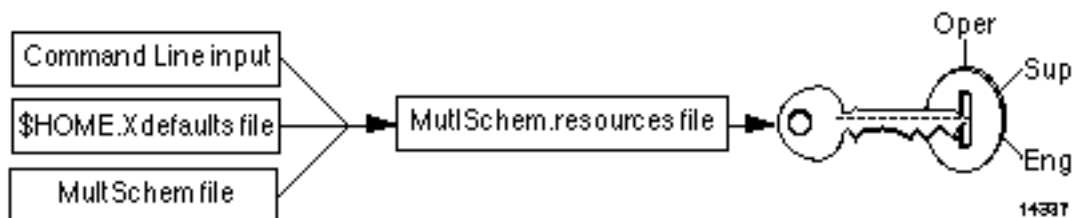
2. Multiple Schematics checks the current keyswitch level of the client U^XS. The keyswitch level must be at the same or lower priority than the requested access level. If it is not, the request is denied.

Attention

Access security applies only at connect time. Once connected, security checks are limited.

Remote server security diagram

A diagram depicting the main remote security procedures follows:



Example of remote security logic

Another example of how remote security logic works is as follows:
If the access level assigned to a request from the remote “ws01” is operatorAccess, but “ws01” appears only in the supervisorAccess access level list, the request is rejected. “ws01” would have to appear in the operatorAccess access level list before the request could be accepted.

Security Table description

The Security Table below indicates whether or not a remote user can invoke a schematic based on the following three parameters:

1. The access level list where the remote is entered.
2. The access level assigned to the remote request.
3. The current keyswitch position of the client U^xS.

Column 1

There are three different access level lists where remotes can be entered:

- engineerAccess
- supervisorAccess
- operatorAccess

The first column of the table, titled “Access level list where the remote is entered” contains four entries, one for each of the three access lists, followed by a default access level (view-only) list for remotes not entered in one of the three defined access level lists.

Column 2

There are four different access levels that can be assigned to a remote request:

- scEngineer
- scSupervisor
- scOperator
- scViewonly

The second column of the table, “Access level assigned to the request,” contains four entries (one for each of the four access levels) for each access level list appearing in column 1.

Column 3

In addition to being used for security checking, the U^xS keyswitch also provides the resultant user access level. There are three possible U^xS keyswitch positions:

- Engineer
- Supervisor
- Operator

The third column, “U^xS keyswitch position,” contains three entries (one for each of the three keyswitch positions) for each access level appearing in column 2.

Column 4

The data in the first three columns is used to determine whether or not a remote user can invoke a schematic. The fourth column, “Does schematic appear on the remote?,” contains either a “yes” or “no” answer for all possible combinations of the parameter values listed in columns 1, 2, and 3.

How to use the Security Table

From line 1 of the table, for a remote entered in the engineerAccess access level list in the MultSchem.resources file, when

1. The access level assigned to the remote request is scEngineer, and
2. The U^xS keyswitch position is Engineer, then
3. The schematic *will* appear on the remote node.

From line 4 of the table, when

1. The access level assigned to the remote request is scSupervisor (not scEngineer), and
2. The U^xS keyswitch position is Engineer (unchanged), then
3. The schematic *will not* appear on the remote node.

Attention

It should be noted that any remote user who has user id and password authorization can connect to the U^xS and get view-only access to the process.

Security Table

The Security Table, which defines whether or not you can display a Multiple Schematic on a remote, is shown below.

Table 3-8 Security Table

Access level list where the remote is entered	Access level assigned to the request	U'S keyswitch position	Does schematic appear on the remote?
engineerAccess	scEngineer	Engineer	yes
		Supervisor	yes
		Operator	yes
	scSupervisor	Engineer	no
		Supervisor	no
		Operator	no
	scOperator	Engineer	no
		Supervisor	no
		Operator	no
	scViewonly	Engineer	yes
		Supervisor	yes
		Operator	yes
supervisorAccess	scEngineer	Engineer	no
		Supervisor	no
		Operator	no
	scSupervisor	Engineer	no
		Supervisor	yes
		Operator	yes
	scOperator	Engineer	no
		Supervisor	no
		Operator	no
	scViewonly	Engineer	yes
		Supervisor	yes
		Operator	yes

Table 3-8 Security Table (continued)

Access level list where the remote is entered	Access level assigned to the request	U'S keyswitch position	Does schematic appear on the remote?
operatorAccess	scEngineer	Engineer	no
		Supervisor	no
		Operator	no
	scSupervisor	Engineer	no
		Supervisor	no
		Operator	no
	scOperator	Engineer	no
		Supervisor	no
		Operator	yes
	scViewonly	Engineer	yes
		Supervisor	yes
		Operator	yes
Remote is not entered in any access level list	scEngineer	Engineer	no
		Supervisor	no
		Operator	no
	scSupervisor	Engineer	no
		Supervisor	no
		Operator	no
	scOperator	Engineer	no
		Supervisor	no
		Operator	no
	scViewonly	Engineer	yes
		Supervisor	yes
		Operator	yes

Access at each security level

A list of the conditions under which access to Multiple Schematics from a remote X-server is granted at each security level is as follows:

Attention

The restrictions described for each security level listed below apply only to *initial* connection requests.

- **Viewonly**—For view only access, a user with user id and password access can request that the Multiple Schematics window be displayed on any X-server, local or remote. The user can only view the process and cannot make any process changes regardless of the keylevel of the station. All requests that would result in process changes are stopped on the X-side, and the user is informed of an access level error.
- **Operator**—For Operator level access, the following conditions must be met for a connection to be made:
 - the user must have login access to the U^xS where Multiple Schematics is installed,
 - the X-server must be authorized for operator access,
 - the current keyswitch level of the U^xS must be Operator, and
 - the user has the same keyswitch level that is currently set at the station.
- **Supervisor**—For Supervisor level access, the following conditions must be met for a connection to be made:
 - the user must have login access to the U^xS where Multiple Schematics is installed,
 - the X-server must be authorized for Supervisor access,
 - the current keyswitch level of the U^xS must be Operator or Supervisor, and
 - the user has the same keyswitch level that is currently set at the station.
- **Engineer**—For engineer level access, the following conditions must be met for a connection to be made:
 - the user must have login access to the U^xS where Multiple Schematics is installed,
 - the X-server must be authorized for Engineer access, and
 - the user has the same keyswitch level that is currently set at the station.

Keyswitch level changes

US and LCN schematic security is based around a single user working at a single station. Because of this paradigm, the single keyswitch level applies to all process interactions originating from that U^xS, whether locally in the native window via Multiple Schematics, or remotely via Multiple Schematics.

For example, *if* a local U^xS user decides to change the keyswitch to Supervisor to perform a specific task, *then* all users of remote Multiple Schematics now also have Supervisor access and capabilities

Parameter changes

On a remote server schematic, the range of LCN parameters whose values can be modified depends on the keyswitch position on the client U^xS. The following table lists all the parameters that can be changed from a remote server for each keyswitch position on the client U^xS.

Table 3-9 Keyswitch Positions and Parameter Changes

Keyswitch Position on U ^x S	Parameter Levels on Remote that Can Be Changed
Operator	Operator
Supervisor	Operator, Supervisor
Engineer	Operator, Supervisor, Engineer

Actors that modify keyswitch access

This class of actors allows a user to configure targets to modify the keyswitch level of a US or U^xS for certain process interactions. The US portion of the U^xS is designed to support only a single user, so it follows that the keyswitch level modification affects subsequent process interaction originating from that U^xS, whether in the native window locally through Multiple Schematics or remotely through Multiple Schematics.

For example:

- A local or remote user decides to use a target to change keyswitch level to Supervisor to perform a specific task
- All users of Multiple Schematics and the native window now also have Supervisor access and capabilities

3.6 Changing the Area Database

Area change impact

If you change the area on a U^xS running Multiple Schematics, the schematics will continue to be displayed. However, security and operational functions for the schematics displayed on the X-side may be impacted by the area change.

You should make area changes on a U^xS running Multiple Schematics only if you and all users of schematics from the Multiple Schematics client are aware of the impact of the change.

For example:

- As shown in the following figure, the local U^xS (Station A) is running Multiple Schematics. The native LCN window is showing, with a schematic being displayed in an X-Window.

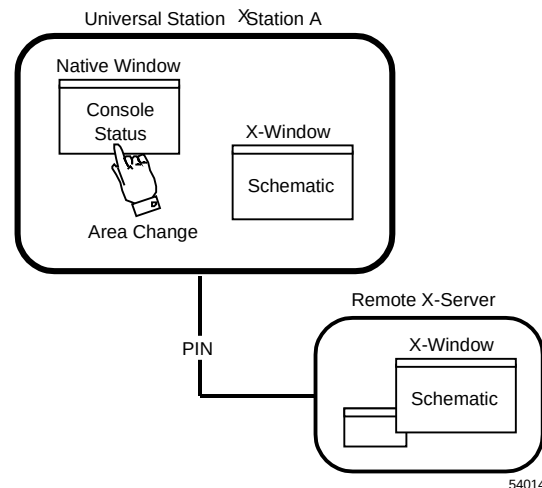


Figure 3-2 X-Window Display of Schematic

- A remote X-server is displaying a schematic that it is receiving from the Multiple Schematics client in Station A.
- On the Console Status display at Station A, a user does an area change. This changes the area database in the station's LCN processor
- The operator at the remote X-server calls up a schematic that is not on a path in the Pathname Catalog for the current area because the area database in Station A has been changed. The following message appears in the schematic window: **Display is Missing or Inaccessible**.
- The operator has no way of knowing why this message appeared, and will have to contact people at Station A and ask what happened.

Advising users of area change impact

Before making an area change, you should advise the users of X-side schematics. Local users are likely to know about the area change and its impact, but remote users probably will not.

After an area change, the following area database entities can have a significant effect on local and remote users of schematics in X-side windows:

- **Unit Assignments**—Users can view points in units not in the new area, but cannot change parameter values in those points. An area change could cause a user to be confused about which points' parameter values can be changed and which cannot. This would not impact a remote user who has no write access to parameter values.
- **Ackstat Collectors**—If the unit of a point in alarm is in the old area but not in the new area, the Ackstat of this point will change from “alarm” to “no alarm.”
- **Alarm Collectors and PRIMMOD Collectors**—Since unit and point assignments could be different, and annunciator names could be different or redefined after an area change, values returned by the Alarm and PRIMMOD collectors could be misleading. It is strongly suggested that schematics containing these collectors be reinvoked following an area change.
- **History Collectors**—Since unit and point assignments could be different after an area change, subsequent history collection requests may return errors.
- **Button File**—If the function keys on the Auxiliary keyboard on the U^xS are configured differently in the new area, pressing the function keys on the remote keyboard will produce a different result.
- **Pathname Catalog**—A schematic that could be requested before an area change, may not appear when requested after the area change. This is because the new pathame catalog does not have a pathname that points to the location of the schematic.

Most of the area database entities define standard displays, logs, reports, and journals. Because standard displays do not appear in X-side windows and because logs, reports and journals can be demanded only from standard displays, only users of the native window are affected.

An alarm annunciation policy is also defined in the area database. Because alarms are annunciated only at the local U^xS, remote users are not affected.

3.7 Schematic Operation in X-Windows

Value update rate

Values appearing in an Mult Schem window may update at rate different than values appearing in the native window. For example, values updating at, say once a minute in the native window, will update once every 2 minutes in a nonfocused Multiple Schematics window with a default update rate of 8 seconds. The formula used to determine the update rate for Multiple Schematic (MS) values is as follows:

$$\text{MS values update rate} = \text{configured set collection value} * \text{focused/unfocused update rate}$$

Actors

Actors in X-Window schematics behave as they do in schematics on a US and in the native window on a U^xS, except in the following situations:

- If an actor calls for one of the standard Honeywell displays, such as a Group display, a Detail display, or the Organizational Summary display, that display will either appear in the native window, or will not appear and an error message appears in the Multiple Schematics window. What happens depends on two things:
 1. whether or not the user is remote, and
 2. the value configured for the useNativeWindow display resource or included in a command line argument.

See *useNativeWindow* in Section 2.6, "Application Resources."

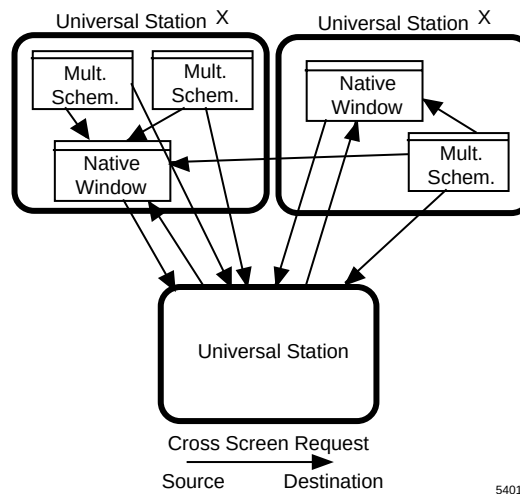
- Actors that accept user inputs to a Text Input Port time out in 100 seconds. On the LCN side, the time out interval for these actors varies from 40 to 120 seconds.

Delay actors

Delay actors suspend further target actions in the window, along with value, conditional, and variant updating.

Cross screen operations

Cross screen requests from a schematic in an X-Window function just as they do from the native window and from a US. However, the cross screen function can only direct a display to the native window on a U^XS or to a screen on a US. This is illustrated in the following figure.



When a U^XS receives a cross screen request from another station, it either shows the requested display in its native window, or an error message is returned to the display that sent the request. This depends on two things:

1. Whether or not the user is remote, and
2. The value configured for display resource `useNativeWindow`.

See `useNativeWindow` in Section 2.6, "Application Resources."

Change zones

Change zones in X-Windows behave as they do in the native window and on USs. This includes typing of data in the Text Input Ports in a change zone, except that in the initial release of Multiple Schematics, keyboard inputs from remote stations are not accepted.

Trend displays

Trend displays in X-Windows behave as they do in the native window and on USs.

Date and time on remote stations

The date and time shown in schematic windows on remote stations are those of the client running on a U^xS. If the client should be in a different time zone, the date and time may not be the same as local time.

Fast Update feature

The Fast Update feature functions only on USs and in the native window on U^xSs.

Formatted tag names

On an LCN-based system, in System Wide Values under Network Configuration, you can configure the system to accept either short (8- character) or long (16- character) tag names.

Users can also configure a ***Formatting On Long Tag Names*** option that causes complete tag name fields to appear on each line, where tag names extend beyond one line. Tag names that appear in Change Zones in Mult Schem windows appear in the formatted form, whether or not the formatting option is selected for the LCN. Tag names that appear in the native window are displayed as specified by the formatting option.

Error handling

Errors are reported as they are for the native window and for schematics on a US. In addition to the types of errors reported on the LCN-side, additional errors related to operation on the X-side are reported.

Honeywell has provided for the possible, future conversion of the Multiple Schematic error messages to languages other than English. For more information, contact your Honeywell representative.

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